HALO: A Heterogeneity-Aware, Language-Conditioned IMU Foundation Model for Label-Efficient Human Activity Recognition

ABSTRACT

Human activity recognition has no foundation model a developer can simply pick up: one pretrained model that reads whatever sensors a device exposes, recognises the application's own activity names, and needs only as much labelled target data as the deployment can afford. Scale alone will not produce one. Inertial signals admit no closed grammar the way language does—a change of mounting, placement, rate, or on-device filtering puts a recording outside any corpus—so compatibility must be built in as inductive bias, not accumulated as coverage. And the usual remedy for open label sets—projecting sensor features into a sentence-embedding space—presumes that sensor similarity tracks linguistic similarity; we measure that it does not, and that it fails in both directions. We present HALO[1], which answers each obstacle with a named mechanism. A *physical-Hz tokenizer* reads every sensor on a frequency axis fixed in Hz, with no resampling anywhere in the path, so the representation is rate-invariant by construction; *acquisition-bounded observability masks* state what a configuration could not measure instead of fabricating it; *per-sensor language conditioning* leaves channel count and ordering free while staying too coarse to fingerprint a corpus; and *evidence retrieval in sensor space* decides among the label strings a deployment declares at run time, with no sensor-to-text projection at all. Pretraining uses no activity labels, and the memory is append-only, so one frozen model spans an entire *label-efficiency curve*—from no target data to full supervision, with no gradient update at any point—rather than a zero-shot-versus-fine-tuning dichotomy. We train on 12 public datasets (20 sensor streams) and evaluate on 7 held-out datasets under a leakage-free, subject-disjoint protocol scored against each dataset's own label strings.

1 INTRODUCTION

A physiotherapist discharges a patient after knee surgery with six prescribed exercises, and wants to know whether they were done. The patient already owns a phone and a watch; the six exercises were chosen for this patient from a long tail of clinical options; and nobody is going to collect and annotate a training set. This is the ordinary case for inertial sensing, and it is not what the field builds for. Almost every deployed IMU system is constructed the other way round: fix the sensor layout, fix the label set, collect a bespoke annotated corpus, train a classifier for it. That pipeline works, and it has to be paid for again from scratch for the next clinic, the next exercise list, and the next phone model.

Three properties of that situation make it hard, and each belongs to the *deployment* rather than to the recognition problem. The label set is **open**—defined per patient, per workplace, per study, and containing activities no public corpus has recorded. The sensing is **heterogeneous**—whatever hardware the user owns, worn wherever they put it, sampled at whatever rate the firmware chooses. And the supervision budget is **small and fixed**—a clinician can demonstrate an exercise a handful of times, not label five hundred repetitions. Human activity recognition (HAR) has no foundation model meeting all three at once: one pretrained model a developer picks up, points at whatever sensors a device exposes, tells the application's own activity names, and asks only for as much labelled data as the deployment can afford. We use *foundation model* in its usual sense—a model trained on broad data, typically self-supervised, producing general embeddings adaptable out of the box to downstream tasks. Adaptation from a handful of labelled examples is therefore within scope, not a departure from it: the question is not whether a deployment supplies labels but how few it must supply. Section 3.1 argues these demands from concrete deployments.

The field's dominant answer has been scale. We argue that scale alone cannot deliver this, for two reasons specific to inertial sensing, and that those reasons dictate two design choices rather than a bigger corpus.

(L1) IMU has no closed grammar—only explicit physics. Language and vision foundation models work because their substrate is, in a real sense, covered by a large corpus: text is generated by a grammar over a finite lexicon, and natural images obey a stable projective geometry, so enough data does approach the space of possible inputs. Inertial signals have no comparable closure. The same motion, from the same body, becomes a different signal if the device is rotated, worn on a different limb, sampled at a different rate, low-pass filtered by an unknown firmware stage, or reported as gravity-inclusive rather than gravity-removed

[1]Source code and pretrained models will be released upon acceptance.

acceleration—not corner cases, but what changes when an app ships on a second phone model. A larger corpus reduces the *probability* of surprise but cannot close the space, because the transformations are continuous and compose. Compatibility must therefore be built in as *inductive bias*—a quantity to be *calibrated*, not maximised. Assume too much (a fixed channel layout and rate) and the encoder cannot read a device it was not built for; assume too little (every scalar channel an independent series carrying a free-text name) and nothing stops the model from using that text as a dataset identifier.

(L2) Sensor similarity does not track linguistic similarity. The standard remedy for open-ended label sets is to project sensor features into a sentence embedding space and retrieve the nearest label string [42, 52, 78]. This presumes the two geometries agree. They do not, and they disagree in both directions. *Drinking coffee* and *drinking water* are the same wrist trajectory but only 0.56 apart in text; *sitting* and *sitting down*—a static posture and a dynamic transition—sit at 0.93. Across 14 label pairs measured with the frozen sentence encoder the field uses (Section 3.3), 38 of 49 cross-comparisons are *inverted.* A single projection asked to reconcile these geometries is mis-specified, and its cheapest way to reduce training loss is to exploit whatever *does* predict the label within a corpus—subject, session, device—which is precisely the memorisation that cross-dataset evaluation exists to catch.

(L3) The quantity that matters is label efficiency. These choices are usually argued about as "zero-shot versus fine-tuning," a false dichotomy: a deployment does not choose between zero labels and a full annotation campaign, it has a budget, and the question is what that budget buys. HALO is therefore evaluated on a single *label-efficiency curve*: one frozen model recognises declared labels with no target data, improves as demonstrated examples are appended to memory with no gradient update and nothing leaving the device, and remains available for supervised fitting when labels are plentiful. Zero-shot is the left endpoint of that curve, not a separate claim.

HALO. We present HALO (**H**eterogeneity-**A**ware **L**anguage-conditioned **O**pen-set model), which answers each obstacle with a named mechanism (Table 1 fixes the names). Against L1, a **physical-Hz tokenizer** commits to the physics that is genuinely universal (co-located axes form a rigid sensor; motion energy lives at physical frequencies; gravity is a signed DC offset) and to nothing that varies by deployment: each sensor is read on a frequency axis fixed in Hz by a constant-Q filterbank over a native-rate transform, so the representation is rate-invariant and anti-aliased *by construction*—no resampling exists anywhere in the path—while **acquisition-bounded observability masks** state which bands a given rate and window could actually measure rather than fabricating them. **Per-sensor language conditioning** attaches identity to the *sensor* rather than to each scalar channel, so channel count and ordering are free while the conditioning stays too coarse to fingerprint a corpus. Against L2, **evidence retrieval in sensor space** declines to reconcile the two geometries at all: encoded exemplars are retrieved from an **append-only memory** by *sensor* similarity, and evidence accumulates for whichever label strings the deployment declares at run time. Pretraining is entirely activity-label-free.

We contribute: the *no closed grammar* argument and its realisation as a physical-Hz tokenizer; the direct measurement that sensor and linguistic similarity are frequently *inverted*, and its architectural consequence, an evidence engine that never forces the two geometries through one projection; a reframing of the evaluation as one *label-efficiency curve* under a leakage-free, subject-disjoint cross-dataset protocol scored against each target's own label strings, with a dataset-identity probe for corpus memorisation; and a released checkpoint behind a small `predict()` interface, trained on 12 public datasets spanning 20 sensor streams and 93 labels and evaluated on 7 held-out datasets.

2 RELATED WORK

HALO sits at the intersection of three literatures, and in each some part of what we do already exists. We name the closest work and state what is left.

Cross-dataset and foundation models for IMU sensing. Deep HAR has moved from single-dataset CNN–LSTM models [49] through methods attacking one axis of heterogeneity at a time [50, 51, 73] to self-supervised encoders intended to transfer: LiMU-BERT [74] pretrains by span-masked reconstruction at 20 Hz, CrossHAR [23] adds contrastive learning, and ssl-wearables [80] scales multi-task self-supervision to ~700k person-days of wrist accelerometry. A second wave targets foundation scale, establishing the label-efficiency argument and learning from incomplete recordings [45, 76], learning motion-similarity metrics at $\sim 10^9$ windows [75], and pairing sensor data with captions [86]; IMU foundation models outside HAR pursue the same transfer goal on a different task—Tartan IMU [87] pretrains for inertial positioning across ground, aerial and legged platforms, then adapts per-deployment with LoRA; general time-series and tabular models [20, 47, 83] trade domain bias for breadth, and Inertia-1 [77] runs a controlled lifecycle study over modality, placement, rate, and window length [7]. *What is left:* all of these treat sampling rate as a preprocessing decision to be normalised away. We argue that is where the information loss happens, and Section 5.4 measures it.

Absorbing acquisition heterogeneity: resolution and language. Two lines address the input side. The first, and the closest to our L1 mechanism, is not in HAR: rate-invariant autoencoding [30] quotients out temporal warping, FlowState [21] builds a sampling-rate-*equivariant* forecaster, and universal spectral tokenization [58] ingests native, irregular grids without resampling. Within IMU sensing, MoPFormer [81] tokenizes into learned motion primitives but resamples every dataset to 100 Hz first, so its codebook is defined over a fixed-rate grid. The second describes the sensor in language, and is crowded: GOAT [42] supervises over labels and on-body positions; ZeroHAR [15] conditions on categorical sensor context; CHARM and its successor [16, 17] feed channel-level descriptions into a channel-order-equivariant transformer, finding text conditioning on channel identity among their largest drivers; SensorLLM [35] and oneHAR [70] take related routes; UniMTS [84] instead encodes placement structurally as a joint graph with rotation-invariant augmentation. *What is left, honestly:* "sensor identity as natural language" is not ours to claim, and our differences are narrower than the general idea. FlowState conditions on the rate—it is equivariant, adapting its computation—whereas HALO's features are *anchored* to physical Hz, so no conditioning is required to compare across devices. And the channel-description line [16, 17, 35] attaches free text *per channel*, exactly the configuration whose corpus-identifiability we argue against and test (Table 6); ZeroHAR's attributes are a fixed closed set, whereas free text composes to descriptions never seen.

Retrieval, memory, and few-shot recognition. Deciding by comparison to stored examples is old in this field and new in its current form. The ubicomp lineage recognises gestures by template [25, 36, 46]; the modern one is episodic and parametric [63, 69], and in language and vision attaches non-parametric memory to a frozen model [9, 24, 28, 33, 44]. Memory-modular classification [27] states as its contribution the property we build our recognition stage around: generalise to new classes by replacing memory contents, without retraining. In sensing, RAG-HAR [62] retrieves over signal descriptors and lets an LLM decide; ZARA [34] retrieves multi-sensor evidence into an LLM agent, under the same "evidence-grounded" name we use for a different mechanism; and in-context time-series classifiers [18, 31] collapse zero- and few-shot into one forward pass. *What is left:* not "we retrieve," but that retrieval happens where similarity is *sensor* similarity between encoded exemplar *patches*, from an encoder conditioned on acquisition configuration—which is what lets a wrist query at an unusual rate retrieve comparable evidence, and which descriptor-space, LLM-agent, and CLIP-space retrieval cannot do. To that we add calibrated rejection [19, 57]. Section 5.3 reports the control that decides whether the distinction is real: the same memory mechanism on a competitor's encoder.

What the closest systems claim. Appendix A, Table 7 summarises which axes each addresses. No system addresses all of them, which is true and also the least interesting thing to say—Section 3.3 argues why our set is entailed rather than assembled.

A note on scale. We do not claim the scale-first programme fails on its own terms: on population-scale health outcomes, models trained on tens of millions of hours [45, 80, 86] are the right instrument and we would not beat them. Our position is not "less data, cleverer model" but that a corpus closes only the axes it *samples* (Section 3.2); the axes generated at deployment time call for a different remedy, and are under-served because they are invisible to the benchmark protocols the field uses.

3 MOTIVATION

3.1 Why This Problem: Three Deployments and What They Demand

That IMU-based activity recognition is useful needs no argument. What needs one is why it should be *open-set*, *heterogeneity-aware*, and *label-efficient*—three qualifiers that add real difficulty and that a system could reasonably decline. Our answer is that the deployments motivating this work force all three at once. **Rehabilitation adherence**: the exercise set is chosen per patient from a long clinical tail, the sensor is the patient's own phone or watch worn wherever they choose, and supervision is whatever fits one appointment. **Occupational strain**: a safety team counts one risky motion whose vocabulary belongs to that facility, over a hardware fleet accumulated across years, and collecting a labelled corpus would mean instrumenting the floor with video and consenting every worker in frame—the thing the deployment exists to avoid. A benchmark demands none of the three, which is why a model can lead one and be inapplicable to both of these.

Each qualifier rests on a different argument. **Open-set** is not optional because the label set is not the model's to choose. Nor will more data fix this, because *the annotation channel for inertial data is uniquely poor*. Images and text can be labelled from the artifact itself, which is why annotation crowdsources and those corpora are enormous; nobody can label an accelerometer trace by looking at it. Labels must come from synchronised video, scripted protocols, or self-report—all expensive, all biasing the data toward laboratory conditions. That is the mechanical reason HAR corpora are small and scripted while image corpora are large and natural. **Heterogeneity-awareness** is not optional because its cost *recurs*—per device model, per placement, per firmware revision, for as long as the product ships. A vendor covering fifty device-by-placement combinations either maintains fifty models or ships one for the lowest common denominator; in practice the second happens, which is much of why deployed phone activity recognition is still a handful of coarse ambulatory classes. This is not only our reading: UniMTS [84] ablates its own components and attributes more of its performance to acquisition-configuration handling than to label-text handling. And the **label budget** is the right axis because a system consuming demonstrations *without a training cycle* needs nothing retrained and nothing re-validated, and the demonstration never leaves the device—in clinical and workplace settings a legal precondition, not a convenience.

3.2 Inertial Signals Have No Grammar—But They Have Physics

The differences between configurations are not cosmetic—the same activity, recorded by three of our own sources, shares no sample grid at all (Appendix B, Figure 6). Our corpus alone (Table 2) reaches the ear and head, from smart glasses and earbuds—devices that did not exist as HAR platforms when most public benchmarks were collected—and ranges from one 3-axis wrist accelerometer to six simultaneous placements on one body, which a fixed-width input layer cannot accommodate without padding artifacts or per-dataset heads. Worst are the conventions the data does not announce: two streams that look identical may differ in gravity convention (Android reports total acceleration, iOS `userAcceleration` removes it), in units, or in an unadvertised low-pass filter. In assembling our corpus we found public datasets already transformed—one per-axis min–max normalised, destroying physical scale; another shipping gravity-removed acceleration under a name suggesting total acceleration. A model that silently assumes a convention will be wrong on data it has every reason to think it understands.

This heterogeneity is usually treated as a nuisance to be absorbed by pretraining on more of it. That is the wrong diagnosis, and it changes what the solution has to be. The map from motion to recorded signal is composed at deployment time from a chain of choices—mounting orientation (a continuous $SO(3)$ nuisance), placement, rate, filter chain, gravity convention, sensor suite—and the chain composes. There is no finite enumeration to cover, so a corpus 10× larger reduces the *probability* that a new deployment is far from anything seen without closing the tail—and that tail is cheap to encounter: a second phone model, or a firmware update, is enough.

If compatibility cannot come from coverage it must come from *inductive bias*. This is the standard move, but for inertial sensing it fails on *both* sides. **Too much bias:** layout-locked encoders fix a channel order, count, and rate; they are the most sample-efficient thing to train and cannot read a device they were not built for. Orientation-*invariant* designs are a subtler version of the same error, because invariance is lossy—discarding orientation discards gravity, and with it the distinction between lying down and standing still. **Too little bias:** treating every scalar channel as an independent series identified only by a free-text name is maximally flexible and creates a shortcut, because in a multi-corpus training set that string is nearly a sufficient statistic for *which corpus* a window came from, as are the rate and the channel count. A model rewarded for predicting a label within a corpus can satisfy much of the objective by inferring the corpus and predicting its marginal—and cross-dataset evaluation is where that stops working. We therefore treat corpus identifiability as a measurable failure and probe for it directly (Table 6).

The calibrated position HALO adopts is to commit to the physics that does not vary and to nothing that does. Three things do not vary: co-located axes on one rigid device share a coordinate frame and must be rotated together; motion energy lives at physical frequencies, so a band means the same thing at every rate; and the quasi-static component of an accelerometer is gravity, carrying posture, so it must be preserved as a signed quantity. Everything else is deferred to run time and described in language at the granularity of the *sensor*. This is stronger than resolution *equivariance*, which conditions computation on the rate [21, 58]: an equivariant model must still be told the rate, and its features at two rates are related by a learned transformation rather than being the same quantity. We want a feature to *denote* a physical frequency, so a band from a 20 Hz phone and one from a 100 Hz research IMU are comparable with no conditioning at all—which is what makes them comparable in a shared memory.

3.3 Sensor Similarity Does Not Track Linguistic Similarity

The second obstacle is on the output side. Because deployments name their own activities, a reusable model cannot have a fixed output layer, and the field's answer has been to embed label strings with a frozen sentence encoder and align sensor features to that space, so recognition becomes nearest-neighbour retrieval over an arbitrary label library [42, 52, 78, 84]. The premise is rarely stated: that proximity in sentence-embedding space proxies proximity in motion space. Figure 1 measures it on our own vocabulary and finds it not merely imperfect but inverted.

The failure is structural rather than a deficiency of one sentence encoder. Natural language encodes activities by their *object and purpose*—what is being drunk, what is being cut—while an IMU sees only the *kinematics of the limb*,

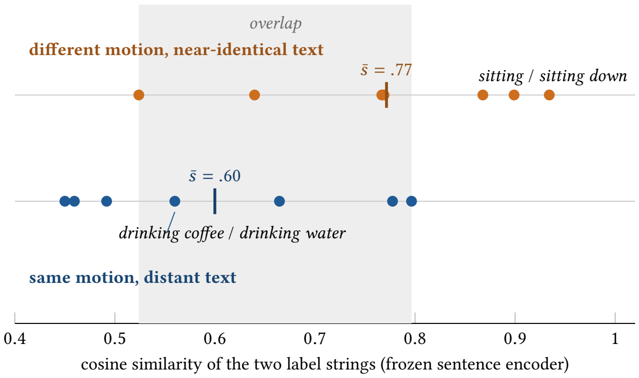

Figure 1: Sensor similarity and linguistic similarity are inverted, not merely noisy. Each dot is one pair of activity names from our label vocabulary, placed at the cosine similarity of their frozen sentence embeddings (all-MiniLM-L6-v2, the encoder language-aligned HAR models use). Pairs in the *top* lane denote physically distinct activities; pairs in the *bottom* lane denote the same physical motion. If text similarity tracked sensor similarity the lanes would separate, with the bottom lane on the right. They separate the wrong way: the mean of the physically *distinct* pairs (0.77) exceeds the mean of the physically *identical* pairs (0.60), and 38 of the 49 cross-lane comparisons are inverted. The motion grouping is our judgement; the similarities are measured. A single projection from sensor space into this text geometry is therefore mis-specified—which is why HALO retrieves in sensor space and votes in label space (Section 4.4).

which is invariant to the object; conversely, morphological relatives (*sit*/*sit down*) share nearly all of their surface form while denoting opposite dynamics. No amount of alignment training repairs a target geometry that is wrong. The consequence is to stop trying: HALO keeps language as the *interface* for naming activities but not as the *metric space* in which recognition happens.

Why these answers are entailed, not assembled. "We combine more components" is not a contribution, and we claim no component as novel. Our claim is that physical-Hz *anchoring* is what makes a single, flat, cross-device memory well-posed in the first place—an *enabling* result rather than an assembly. Such a memory requires that a query encoded on one device and an exemplar encoded on another be comparable *as vectors*, with no per-configuration index to reconcile them; neither resampling nor rate-equivariance provides that, which is why existing retrieval-based sensing systems retrieve somewhere else—descriptor space [62], an LLM agent [34], CLIP space [86]—and why none needs the property we claim. Once it is in view the remaining choices stop looking like options: they are *mutually entailed* (Appendix A). Because "it is just a combination" is an empirical objection, Section 5.4 runs the conventional assembly as a rival arm.

4 DESIGN OF HALO

4.1 System Overview

HALO is organised as two phases with a hard boundary (Figure 2). **Phase A** learns a representation using no activity labels whatsoever; **Phase B** turns that frozen representation into a recogniser for whatever label set a deployment declares, using labels only as annotations on stored exemplars. The boundary is where the argument of Section 3 lands: L1 says compatibility must be structural, so it belongs in the encoder before any label is involved; L2 says the label decision must not be a projection into text space, so it belongs after the encoder, in a component swappable per deployment.

One window, end to end. A deployment hands HALO a 6 s window from whatever sensors it has—say a 50 Hz watch accelerometer and gyroscope, six channels—plus one description string per sensor and its list of activity names. The tokenizer cuts each channel into patches and turns each patch into one token: a 32-band constant-Q readout on a frequency axis fixed in Hz, a signed DC term carrying gravity, and two masks recording which of the 32 bands this rate and patch length could actually resolve. Nothing is resampled, so band k denotes the same physical frequency whether the device ran at 20 or 100 Hz. Each token is then biased by the sentence embedding of its sensor's description, and the token set—unordered over channels, positioned in physical time—enters a 7.17M-parameter transformer that emits one vector per patch. Those vectors are the interface between the phases: Phase A shapes them with objectives that never see a label, and Phase B compares them, by cosine similarity in that same space, against a memory of encoded exemplars, accumulating evidence for whichever activity names were declared. The demands of Section 3.1 map onto this path one-to-one, and Table 1 fixes the name we use for each mechanism throughout. The distinctions those names carry are the contribution: not "a spectral tokenizer" but one anchored to *physical* Hz; not a Nyquist mask but one bounded by the *acquisition* clock; not per-channel text but per-*sensor*; not retrieval in CLIP space but in *sensor* space.

4.2 The Physical-Hz Tokenizer

4.2.1 Per-sensor language conditioning: sensors, not channels. HALO's first commitment is that the unit of description is the *sensor*, not the scalar channel. Three co-located accelerometer axes share a coordinate frame: they rotate together, experience the same gravity vector, and no physically realisable transformation moves one without the others.

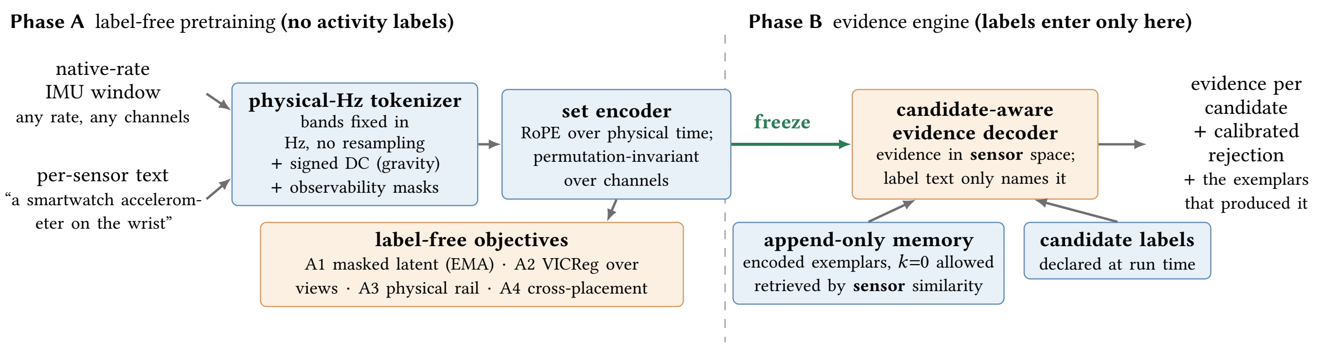

Figure 2: HALO end to end. One left-to-right path. Phase A trains the tokenizer and encoder with objectives that never see an activity label. The encoder is then frozen and crosses the boundary unchanged—it is the *same* weights that encode a query and that encoded every exemplar in memory, which is what lets the two be compared as vectors. Phase B adds no parameters to that path: it appends encoded exemplars to a memory and accumulates evidence for the label strings declared at run time. Adding an activity, a placement, or a device is an append; no gradient step occurs anywhere to the right of freeze.

Table 1: Each deployment demand, the mechanism that answers it, and the name used for that mechanism throughout this paper. The qualifiers are load-bearing: each marks where HALO departs from the nearest existing technique. Rows 1–3 are the tokenizer (Section 4.2) and are what let one weight set accept any rate, channel count, and placement with no architectural change; rows 4–5 are the evidence engine (Section 4.4). Row 3 states a property we *probe* rather than assert (Table 6), and row 5 is why no label budget requires a gradient update.

Demand	Mechanism	Name
Rate and anti-aliasing differ per device	constant-Q filterbank on a fixed physical-frequency axis over a native-rate transform; no resampling anywhere in the path	**physical-Hz tokenizer**
What a device *could* measure differs	per-token masks bounded by the ACQUISITION clock, not the rate the grid happens to be stored at	**acquisition-bounded observability masks**
Channel count, order and placement differ	free-text identity attached to the SENSOR, leaving channel count and order free while staying too coarse to fingerprint a corpus	**per-sensor language conditioning**
The label set belongs to the deployment	retrieve encoded exemplars by SENSOR similarity; accumulate evidence for whatever strings are declared at run time	**evidence retrieval in sensor space**
The label budget is small and fixed	append exemplars to memory; no gradient update at any budget	**append-only memory**

Treating them as independent series discards this and lets per-channel text act as a corpus fingerprint. HALO therefore models a stream as a *set of sensors*, each a triad with its own device, placement, and gravity convention—which also makes the augmentation physics correct, since an $SO(3)$ rotation is applied jointly to a triad and its co-located gyroscope rather than to a single axis.

Accelerometer streams are gravity-aware. The quasi-static component below the analysis band is split off as a *signed* per-axis DC feature rather than removed, so posture survives. On 200 real waist-phone windows each of standing, sitting and lying, the frequency path separates no pair distinctly (all pairwise cosines within 0.84–0.88) while signed DC places lying nearly orthogonal to standing (−0.04); a design achieving orientation robustness through *invariance* discards that quantity, and with it the distinction between lying down and standing still. Figure 7 shows this and its limit—standing and sitting stay at 0.99 in DC, the same pair Section 3.3 finds inseparable in text, so neither path suffices alone. Because gravity conventions differ across platforms, the convention is declared in the sensor's text and a gravity-removal augmentation is applied with low probability, so the encoder sees both conventions for the same motion.

4.2.2 A physical-frequency constant-Q filterbank. Each patch of each channel becomes one token entirely in the physical-frequency domain (Appendix F, Figure 8). The patch is Hann-windowed and its DC component removed, then transformed by a zero-padded real DFT of fixed length S=256. The zero-padding is what makes this rate-agnostic: for a patch sampled at rate r, DFT bin m corresponds to physical frequency $\phi[m] = mr/S$, so the *same* matrix of bin indices maps to different Hz grids at different rates, and the readout can be defined once in Hz. That readout is a constant-Q Gaussian filterbank [10, 11] with K=32 centres log-spaced over $[0.3, 15]$ Hz at Q=4. The limits are physical rather than tuned: gait cadence occupies 0.5–3 Hz, limb and hand motion add harmonics to roughly 10–12 Hz, and content below 0.3 Hz is gravity, which the signed DC feature already carries.

Rate-invariance is therefore a property of the construction rather than of a learned approximation: a 2 Hz gait rhythm deposits energy in the same band whether the device sampled at 20 or 100 Hz, and because no interpolation exists in the path there is no aliasing to introduce and no fabricated sample to explain away (Figure 3). This is the property, argued in Section 3.2, that rate-*equivariance* [21] does not provide and that one flat cross-device memory requires. Appendix F records the DFT- and patch-length guardrails.

4.2.3 *Observability: say what could not be measured.*

Rate-invariance would be dishonest without a second mechanism. A 20 Hz stream cannot report energy at 12 Hz, and a 0.4 s patch cannot resolve a 0.3 Hz band. Rather than emitting a small number the encoder might read as "no energy here," each token carries two masks. The *Nyquist observability* mask marks band k (centre c_k, width $\sigma_k = c_k/2Q$) observable only if $c_k + 2\sigma_k \leq 0.9\,(r/2)$, a 10% guard so a band's Gaussian tail does not straddle the aliasing edge; the *resolution* mask marks it resolved only if one full cycle fits in the patch, $c_k \geq 1/D$ (Figure 8). A band that is present but blurry is flagged, not zeroed, and streams resampled by their publisher take the Nyquist bound from the *acquisition* rate, since upsampling cannot create information.

4.2.4 *Factored language conditioning.*

Tokens from a weight-shared tokenizer are agnostic to what produced them, so acquisition context must be injected. HALO factors it in two: a **role** string per channel naming *only* the axis (`"x"`, `"y"`, `"z"`), constant across the corpus; and a **sensor identity** string per sensor carrying device, modality, placement and gravity convention, e.g. `"a smartwatch accelerometer on the wrist; includes gravity"`. Both are embedded by the same frozen sentence encoder [53] and fused into the token, the sensor embedding through a gated residual initialised near zero so conditioning starts as a gentle bias and grows only if it earns its place.

The factorisation is the point: the description a channel receives is shared by every sensor of that kind across the corpus, which is what makes it too coarse to identify a dataset. Numeric metadata is deliberately *absent*, because a frozen sentence encoder represents numbers unreliably; rate and duration enter through the bin→Hz map and the masks instead. Sensor strings are paraphrased during training and sometimes dropped to a generic one, so the encoder must tolerate an unfamiliar description at deployment. Whether this buys the anti-memorisation it is designed for is empirical: Table 6 tests it against per-channel-text and layout-locked variants.

4.2.5 *Set encoder over physical time.*

Conditioned tokens enter a dual-branch transformer [67] alternating *temporal* self-attention within a channel and *cross-channel* self-attention within a patch, with channel masks so absent or padded channels never contribute. Channels carry no positional index at all—their identity is text—so the encoder is permutation- and count-invariant over channels by construction, and a device with three channels and one with twelve are the same kind of input. Temporal positions use rotary embeddings [65] keyed to *physical time* rather than patch index: the filterbank's counterpart on the time axis, so a 1.2 s gesture receives the same relative code regardless of rate or patch length. The encoder is 7.17M trainable parameters (d=256, 6 layers, 8 heads) plus a frozen all-MiniLM-L6-v2 sentence encoder (22.7M) shared by sensor conditioning and label text, reading 6 s windows tokenised at two resolutions. Because rotary positions are relative and the tokenizer uses no cross-patch statistics, offline and causal operation differ only by the attention mask and the encoder implements both—an affordance, not a claim, since we do not measure a causal arm here.

4.3 Phase A: Label-Free Pretraining of the Physical-Hz Tokenizer and Encoder

Phase A trains the tokenizer and encoder with objectives that never observe an activity label. This is a change of principle, not an economy: as long as the objective is closed-vocabulary label prediction, the model is rewarded for exploiting whatever predicts labels within the training corpora—the failure of Section 3.2—and removing labels also unlocks unlabelled data, which is where the scale in wearables is (full spec in Appendix G).

(A1) Masked prediction, with two targets. A validity-aware mask over physical-time intervals hides a subset of tokens; at masked positions the encoder predicts both the *fixed* filterbank features of the hidden content and the *contextual latents* a stop-gradient EMA teacher [2, 3, 22] produces from the unmasked window. The fixed target grounds the representation in physical quantities; the latent target lets it become more abstract than its own input features.

(A2) VICReg over augmented views. Two independently augmented views are pulled together under VICReg [5] rather than a contrastive loss, for a reason specific to this corpus: NT-Xent declares the other windows in a batch negatives, and in a multi-dataset corpus the batch is full of the same activity from other subjects and datasets, so it actively pushes apart things that should be together. VICReg needs no negatives; SimCLR [14] and SupCon [29] remain selectable as controls. Text augmentation uses a *shared* seed across views, so a pair never differs only in whether its configuration was described.

(A3–A5) Three low-weight rails complete the objective: time–frequency consistency against an auxiliary time-domain encoder discarded at inference (inspired by TF-C [85] but not faithful to it); a cross-placement term over *verified*

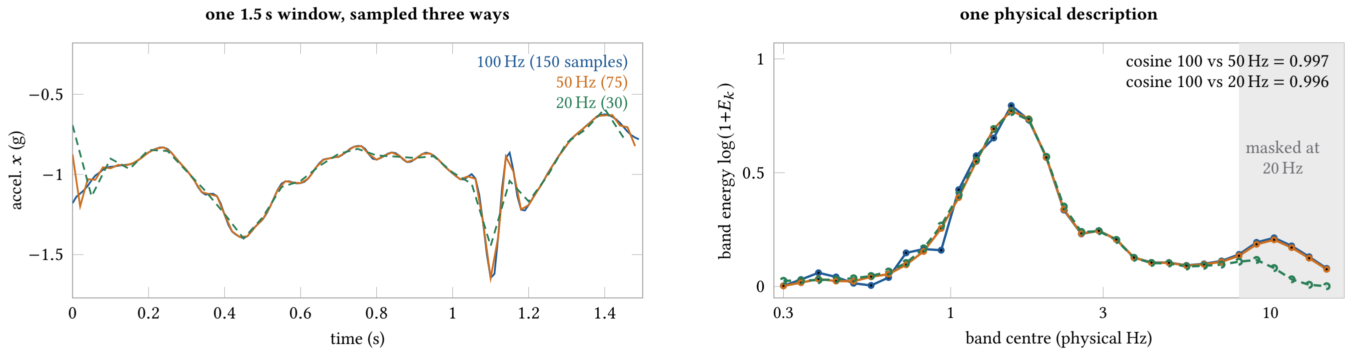

Figure 3: Rate-invariance, measured rather than argued. One 1.5 s wrist patch of *walking* (NFI-FARED) anti-alias resampled to three rates and run through the tokenizer at each. *Left:* as arrays these are three different objects—150, 75 and 30 numbers, no sample coinciding—so any encoder positioned by timestep index sees three different structures. *Right:* the band energies coincide (cosine 0.997 and 0.996 over the 26 commonly observable bands). The shaded region is what 20 Hz physically cannot see: those bands are *masked*, not estimated. The 20 Hz curve does fall away inside it—exactly the loss an encoder that resampled the whole corpus would silently take on every stream.

simultaneous recordings of one physical event at two or more placements (17,387 paired events), whose positive identity comes from explicit event identifiers written during conversion, never from array alignment or an activity label; and two self-derived targets as a sanity anchor. Model selection uses subject-disjoint k-NN accuracy—labels are a *metric* here, never in the loss.

4.4 Phase B: Evidence Retrieval in Sensor Space

The encoder is frozen. What remains is to turn its representation into a decision over label strings supplied at run time, without the single sensor-to-text projection Section 3.3 showed to be mis-specified. Figure 4 shows the mechanism.

Memory and retrieval. Every labelled window available—from the training corpus and, optionally, the target deployment—is encoded once and written to a versioned table storing the patch vector with its label, subject, dataset, acquisition configuration, and source event. The table is append-only: adding a class, a placement, or a device is a write, not a training run. Every artifact is bound to the Phase-A checkpoint it was built from, and a behavioural probe rejects a bank whose encoder has since changed—silently stale memory is the most dangerous failure mode this design has. Each query patch then retrieves independently in *sensor* space under learned projections, each defining its own subspace and evidence contribution. During training the query's own window, event, and subject are excluded, and contributions are capped per source window and per label so one long recording cannot dominate a vote.

Candidate-aware decoding. Retrieved evidence is not simply averaged. A small set transformer contextualises query and evidence tokens, refines each evidence item's label text as a residual in frozen sentence space—which is what lets a neighbour labelled *walking* but retrieved in a stair-climbing context contribute as *walking upstairs*—and pools

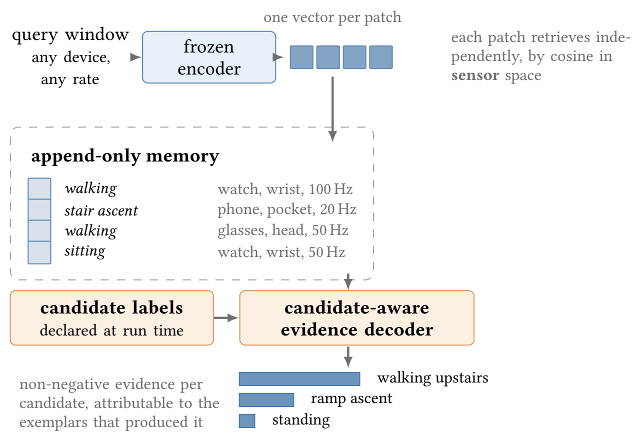

Figure 4: How a decision is made without a sensor-to-text projection. Each patch of the query retrieves independently, by cosine similarity in *sensor* space, from one flat memory whose rows may come from any device—that is what physical-Hz anchoring buys, and it is what a resampling or rate-conditioned encoder cannot offer without a per-configuration index. Label text enters only twice, and never as a metric: to name what a neighbour was, and to enumerate what the deployment will accept as an answer. The decoder emits non-negative evidence per candidate, attributable to the exemplars that produced it. Structural diagram—the bar lengths are illustrative, not measured.

evidence as a residual on the retrieval prior. Candidate tokens carry frozen label text and attend permutation-equivariantly over the evidence, so neither candidate nor evidence order carries meaning. Every sublayer is gated by a LayerScale initialised near zero and the residuals are zero-initialised, so at initialisation the decoder is *exactly* the untrained retrieval-plus-text-ensemble mechanism: every trained arm has an exact untrained control, reported side by side.

Confidence and rejection. There is no UNKNOWN candidate; a rejection confidence is estimated separately [19, 57] from evidence mass, retrieval density, agreement, and the leading margin, calibrated on truth-absent episodes with risk/coverage reported—and truth-present episodes require support, so always-reject is not a solution. Complete label *families* are held out of decoder training for model selection (Appendix I).

Deployment. A deployment is two pieces of configuration and no code: one description string per sensor and a list of activity names, both embedded once and cached, so run-time cost is one encoder pass plus a retrieval. The label-efficiency dial is the memory: at k=0 evidence comes from corpus exemplars and candidate strings alone; at k>0 each demonstration costs one forward pass and is immediately available—no gradient step, no revalidation, no requirement that data leave the device. We expose this behind a small `predict()` interface and package the encoder for Core ML (Section 5.6).

5 EXPERIMENTS

[TO BE COMPLETED] *Corpus, protocol, and table structure are final; orange cells are reserved for results.*

5.1 Methodology

We ask four questions. **(Q1)** Along the label-efficiency curve, where does HALO stand against strong frozen encoders, and is there a regime it owns? **(Q2)** Does the tokenizer's inductive bias do what Section 3.2 claims—and, decisively, does anchoring let one flat memory serve devices it was not built from? **(Q3)** Which Phase-A objectives carry the transfer? **(Q4)** Does the system run on a phone?

5.1.1 Data. We train on 12 public HAR datasets contributing 20 distinct sensor streams and evaluate on 7 never seen during training (Table 2). After conversion to native-rate grids and removal of 95 windows exceeding physically possible sensor rails, the corpus is 1,728,787 non-overlapping windows—2,858 hours—over 93 labels, split subject-disjointly into 300,231 training and 42,909 validation windows, the latter built by a greedy label-cover procedure so all 93 labels are represented. The held-out sets carry 6 to 13 labels each. Appendix B plots the corpus by placement and rate and records the six datasets deliberately excluded and why.

Table 2: The corpus. *#Str* is the number of distinct sensor streams (one per device placement); *Ch* the number of core channels reaching the model (3 = accelerometer only). No two training sources agree on rate, channel set, and placement at once, which is the condition the tokenizer exists to handle. InclusiveHAR comprises 10 able-bodied and 10 physically disabled subjects; TNDA-HAR is distributed without per-window subject identifiers (Section 5.1.3).

Dataset	Hz	#Str	Ch	#Win	Placement(s)
Training corpus (12 datasets, 20 streams)					
CAPTURE-24 [13]	100	1	3	1,530,792	wrist, free-living
WISDM [71]	20	2	6	70,201	pocket + wrist
XRF V2 [32]	50	6	6	34,764	pockets, wrists, glasses
NFI-FARED [41]	100	2	6	26,520	wrist + lower back
HARMES [12]	50	1	6	18,576	wrist
UniMiB SHAR [43]	50	1	3	11,771	trouser pocket
UCI-HAR [1]	50	1	6	10,299	waist
HHAR [64]	50	1	6	9,587	waist
KU-HAR [61]	100	1	6	9,233	waist
SP-SW-HAR [40]	100	2	6	3,757	pocket + wrist
PAMAP2 [54]	100	1	6	3,186	wrist
MHEALTH [4]	50	1	6	1,104	wrist
Held-out evaluation (7 datasets, never trained on)					
RealWorld [66]	50	1	6	11,237	waist
MotionSense [39]	50	1	6	4,534	trouser pocket
USC-HAD [82]	100	1	6	4,350	hip
UT-Complex [60]	50	1	6	3,900	wrist
TNDA-HAR [79]	50	1	6	3,353	wrist
Shoaib [59]	50	1	6	2,100	5 body positions
InclusiveHAR [26]	50	1	6	1,260	waist, mixed abilities

5.1.2 Baselines and the fairness contract. Cross-model comparison here is easy to get wrong, because the models do not accept the same inputs. We state one contract and apply it uniformly (Appendix C): each model occupies a *level* on each of four tiers—architecture, rate, channels/placement, and label vocabulary—and we never modify a released checkpoint's input contract, because doing so discards the model being tested. CrossHAR and LiMU-BERT release no checkpoint compatible with our corpus, so we pretrain them on *our* data with their own published objectives; harnet, UniMTS and NormWear are used as released, which is faithful but means their pretraining corpora differ from ours and from each other. This distinction is material—harnet is pretrained on roughly 700k person-days against our 2,858 hours—and we state it wherever a comparison could be read as data-matched. Closed-vocabulary models are bridged to arbitrary label strings with ConSE [48], so "can predict an unseen label" is available to every model and is not something HALO is credited for.

5.1.3 Protocol. *Scoring.* Following the zero-shot evaluation literature [72], every held-out dataset is scored against its *own* label strings, never a shared training-wide ontology, so a prediction counts only if it retrieves that dataset's exact

Table 3: Per-dataset transfer, grouped by the fraction of that dataset's label vocabulary absent from the 93-label training corpus (*unseen*). Every dataset is scored against its *own* label strings. InclusiveHAR is stratified by ability, because a model that works only for able-bodied users is not a foundation for rehabilitation applications. § TNDA-HAR carries no subject identifiers, so it sits outside the subject-disjoint mean.

		k=0		k=10	
Dataset	unseen	Acc	F1	Acc	F1
Vocabulary fully covered by training					
MotionSense	0%				
RealWorld	0%				
Shoaib	0%				
Vocabulary partly unseen					
InclusiveHAR	33%				
able-bodied					
disabled					
UT-Complex	38%				
USC-HAD	42%				
Mean (subject-disjoint)	23%				
TNDA-HAR§	25%				

activity name. Macro-F1 is primary, with accuracy alongside, because held-out label distributions are imbalanced.

Splits. All few-shot fits and memory writes use subjects disjoint from those evaluated, rotating over grouped folds with a confidence interval where one split would be unstable. TNDA-HAR ships no per-window subject identifiers, so a subject-disjoint boundary cannot be constructed for it; we report it separately and exclude it from the subject-disjoint mean rather than implying a guarantee we cannot honour.

Label leakage. The inference-time label-text ensemble is purged of synonyms drawn from the evaluation datasets, and retrieval hyper-parameters are selected on held-out *training* configurations.

Parity control. To separate the architecture from its native-rate and rich-text inputs, HALO is also evaluated under baseline-equivalent inputs—20 Hz resampling, neutral sensor description—against the best member of each baseline group (Appendix D).

5.2 Per-Dataset Zero-Shot and 10-Shot Transfer

We report the conventional view first: every held-out dataset, scored against its own label strings, at the two ends of the enrollment regime. Datasets are grouped by how much of their vocabulary the training corpus never saw, because that fraction is what the open-set claim is actually about, and it is far from uniform: three of the seven held-out sets are covered entirely by the 93-label training vocabulary, four are not.

Table 4: Macro-F1 versus label budget, averaged over the held-out datasets, subject-disjoint throughout. k=0 is the zero-shot point and k=full the supervised one; they are ends of one curve, not separate experiments, and every row uses *one* frozen encoder across all k. The second header row converts k into the mean fraction of a target dataset it represents. † The mechanism-versus-representation control: if retrieval helps only on HALO features the contribution is the encoder; if it helps equally on harnet's, it is the memory. We pre-register that we report this control regardless of which way it falls.

	labelled target windows per class (k)					
Mechanism	0	5	10	25	100	full
(mean % of target)	*0*	*1.3*	*2.7*	*6.6*	*27*	*100*
HALO memory (no gradient step)						
HALO frozen + probe						
harnet frozen + probe						
harnet frozen + memory†						
UniMTS (native text)						
CrossHAR + ConSE						
LiMU-BERT + ConSE						
NormWear (native text)						

The unseen fractions are properties of the corpus, not of any model: across the seven held-out sets, 14 of 60 label strings (23%) never appear in training. The examples are the point. UT-Complex asks for *drinking coffee*, *eating* and *smoking*—wrist activities a closed sit/stand/walk vocabulary cannot express at all—and InclusiveHAR distinguishes *ramp ascent* from *walking upstairs*. Where a held-out set *is* covered by the training vocabulary, we say so rather than counting it as evidence for an open-set claim it does not test.

5.3 Q1: The Label-Efficiency Curve

This is the headline experiment (Table 4). At each budget k of labelled target windows per class we compare three mechanisms on identical data: HALO's memory, given the exemplars as an append with *no gradient update*; a probe on HALO's frozen features; and the same probe on frozen harnet features. We additionally report *retrieval over harnet features*—the same mechanism on a different encoder—because without it a win could be credited to the mechanism when it belongs to the representation, or the reverse.

The budget is reported in two units, because neither alone is honest: per-class k is what a deployment controls, while the fraction of the target dataset makes the numbers comparable to the 1%/5%/10% convention of the transfer literature. On our held-out sets the two coincide closely (k=10 per class is 0.7–4.8% of a dataset), and the grid stops at k=100 because beyond it the budget exceeds what the smaller sets contain.

Pre-registered hypothesis and kill criterion. We expect retrieval to win at small k, where a nearest-neighbour decision beats a linear fit on a handful of points, and the probe to overtake somewhere between k=10 and k=25—roughly the 3–7% band. If that holds, the regime HALO owns is $k \leq 10$, which is the enrollment setting a user can perform in under a minute. If the probe dominates at *every* k including the smallest, retrieval has no regime of its own; we will report that outcome plainly. Table 3 gives the per-dataset breakdown behind this curve, including InclusiveHAR stratified by ability.

5.4 Q2: Does the Inductive Bias Do What It Claims?

Section 3.2 made two falsifiable claims about the tokenizer. Both are tested directly rather than inferred from end-task accuracy.

Rate-invariance is measured, not assumed. A closed-form check confirms a pure tone deposits identical band energies at 20, 50, and 100 Hz. The end-task version is Table 6: the same encoder on the same data but reading a common 20 Hz resampling of every stream, so the gap measures what resampling costs.

Does anchoring actually make one cross-device memory well-posed? A flat memory shared across devices is only meaningful if a query encoded from one configuration retrieves useful neighbours stored from a *different* one, which is what Table 5 measures. The measurement is unusually clean because parts of our corpus contain *verified simultaneous* recordings—same person, activity, and instant, two or more devices—so querying one stream against a memory built only from its simultaneous partner isolates configuration from subject, session, and activity entirely. We report the same matrix for a resample-to-common-rate variant of our own encoder and for the frozen baselines, so the comparison is of mechanisms rather than corpora.

Is the naive assembly enough? Table 6 includes the obvious alternative as a row: resample every stream to a common rate, attach free text per channel, same retrieval on top. If that arm matches HALO the architecture is a combination and we will say so; mutual entailment predicts it will not, and predicts a reason—per-channel conditioning showing up as elevated dataset identifiability, resampling as a rate gap.

Conditioning must not be a corpus fingerprint. The right-hand columns of Table 6 report a dataset-identity probe: a linear classifier fitted on frozen embeddings of held-out subjects to predict which *source dataset* a window came from. A representation that has absorbed the acquisition description as a corpus identifier makes this easy; one whose conditioning is genuinely compositional does not. It is the only direct evidence we have for the "too little bias" failure mode, so we report it whether or not it favours us.

Table 5: Cross-configuration retrieval: the enabling claim. Each row queries with one acquisition configuration against a memory containing *only* a different one. Because these pairs are verified simultaneous recordings, the only thing that varies is the configuration, so off-diagonal degradation is attributable to it alone. *Retention* is cross divided by same: features that denote the same physical quantity across devices should retain most of their quality; one that resamples or conditions on the rate should not.

	HALO		resample-to-20 Hz	
Query → memory	same	**cross**	same	**cross**
wrist → pocket (SP-SW-HAR)				
wrist → lower back (NFI)				
pocket → watch (WISDM)				
wrist → ear/glasses (XRF V2)				
100 Hz → 20 Hz memory				
cross/same retention				

Table 6: Tokenizer inductive-bias ablation (macro-F1 over the held-out datasets), each component removed independently, with the dataset-identity probe alongside. The *naive assembly* row is not an ablation but a *rival*: the same retrieval mechanism over an encoder built the conventional way. It is the arm that tests whether HALO is a combination of known parts. ‡ *Rate gap* is performance on the 100 Hz held-out cell minus the same data downsampled to 20 Hz—what resampling costs. *ds-ID* is dataset-identity accuracy; lower is better.

	transfer		probe	
Tokenizer configuration	k=0	k=10	rate gap‡	ds-ID ↓
Physical-Hz, sensor-grouped, factored text				
– filterbank (resample to 20 Hz)				
– sensor grouping (per-chan. text)				
– language conditioning (locked)				
– observability / resolution masks				
– signed DC (gravity) feature				
naive: resample + per-chan. text				
harnet [80] / UniMTS [84]			–	
chance	–	–	–	

5.5 Q3: Which Phase-A Objectives Carry the Transfer?

Each Phase-A term is removed independently with all others held fixed (Appendix H). Two controls matter more than the ablation itself. The *untrained control*: the decoder is identity-at-initialisation by construction, so every trained arm is reported against the same architecture untrained, and a component that does not beat its own initialisation is reported as such. The *label-supervised control*: the original supervised recipe remains selectable, so the claim that going label-free helps is measured, not asserted.

5.6 Q4: On-Device Cost and a Real Deployment

We implement the full pipeline as an iOS application and will report encoder latency and peak memory under Core ML, retrieval latency against bank size, and a case study in which a user enrolls activities by demonstration, changes body placement mid-session, and introduces names never enrolled—the deployment form of the label-efficiency argument.

[TO BE COMPLETED] *On-device latency, peak memory, retrieval cost versus bank size, and the case-study figure are added here once the encoder is compiled and profiled.*

6 CONCLUSION AND DISCUSSION

We argued that the obstacle to a reusable IMU foundation model is not a shortage of data but two structural facts about the modality: inertial signals admit no closed grammar, so cross-configuration compatibility must be built in as calibrated inductive bias rather than accumulated as coverage; and sensor similarity does not track linguistic similarity—on our own vocabulary the ordering is inverted—making a sentence-embedding projection a mis-specified target that rewards dataset memorisation. HALO answers with a tokenizer committing to the physics that does not vary, and an evidence engine that retrieves in sensor space and votes in label space.

What we are not claiming. Accepting a variable number or order of channels is not a contribution—padding with a mask is one line of preprocessing—and recognising unseen label *strings* is not one either, which is why every closed-vocabulary baseline gets ConSE. Where labels are plentiful and the configuration is fixed and known in advance, we do not expect retrieval to beat supervised fitting. Our corpus is 2,858 hours, orders of magnitude smaller than the largest released accelerometer models; the evidence engine's ceiling is its encoder, since no decoder recovers from retrieving the wrong activity; windows are single-activity, so the model recognises segments rather than finding boundaries; and the encoder supports causal operation but this submission does not measure it. The most valuable direction is the one Phase A makes free: needing no labels, the corpus can grow into unlabelled free-living data. The label-efficiency curve also carries a pre-registered kill criterion: if a linear probe on frozen features dominates retrieval at *every* budget, the non-parametric mechanism has no regime of its own, leaving only operational properties—append-only updates, on-device enrollment, inspectable decisions—and we would report that rather than reframe the metric.

REFERENCES

[1] Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra, and Jorge Luis Reyes-Ortiz. 2013. A Public Domain Dataset for Human Activity Recognition Using Smartphones. In *Proceedings of the European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*. Bruges, Belgium, 437–442.

[2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. 2023. Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, Vancouver, Canada, 15619–15629. https://doi.org/10.1109/CVPR52729.2023.01499

[3] Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jiatao Gu, and Michael Auli. 2022. data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*. PMLR, Baltimore, MD, 1298–1312.

[4] Oresti Banos, Rafael Garcia, Juan A. Holgado-Terriza, Miguel Damas, Hector Pomares, Ignacio Rojas, Alejandro Saez, and Claudia Villalonga. 2014. mHealthDroid: A Novel Framework for Agile Development of Mobile Health Applications. In *Proceedings of the 6th International Work-conference on Ambient Assisted Living and Daily Activities (IWAAL) (LNCS)*, Vol. 8868. Springer, Belfast, UK, 91–98. https://doi.org/10.1007/978-3-319-13105-4_14

[5] Adrien Bardes, Jean Ponce, and Yann LeCun. 2022. VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. In *The Tenth International Conference on Learning Representations (ICLR)*. OpenReview.net, Virtual.

[6] Billur Barshan and Murat Cihan Yüksek. 2014. Recognizing Daily and Sports Activities in Two Open Source Machine Learning Environments Using Body-Worn Sensor Units. *Comput. J.* 57, 11 (2014), 1649–1667. https://doi.org/10.1093/comjnl/bxt075

[7] Sizhen Bian, Mengxi Liu, Lala Shakti Swarup Ray, Bo Zhou, Bin Guo, Zhiwen Yu, Thomas Plötz, Paul Lukowicz, Siyu Yuan, and Vitor Fortes Rey. 2026. Foundation Models Defining a New Era in Sensor-based Human Activity Recognition: A Survey and Outlook. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* 10, 2, Article 31 (2026), 38 pages. https://doi.org/10.1145/3810230 arXiv:2604.02711.

[8] Sizhen Bian and Paul Lukowicz. 2025. RecGym: Gym Workouts Recognition Dataset with IMU and Capacitive Sensor. UCI Machine Learning Repository. (2025). https://doi.org/10.24432/C5PW4K https://archive.ics.uci.edu/dataset/1128.

[9] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego de Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, Andy Brock, Michela Paganini, Geoffrey Irving, Oriol Vinyals, Simon Osindero, Karen Simonyan, Jack W. Rae, Erich Elsen, and Laurent Sifre. 2022. Improving Language Models by Retrieving from Trillions of Tokens. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*. PMLR, Baltimore, MD, 2206–2240.

[10] Judith C. Brown. 1991. Calculation of a Constant Q Spectral Transform. *The Journal of the Acoustical Society of America* 89, 1 (1991), 425–434. https://doi.org/10.1121/1.400476

[11] Judith C. Brown and Miller S. Puckette. 1992. An Efficient Algorithm for the Calculation of a Constant Q Transform. *The Journal of the Acoustical Society of America* 92, 5 (1992), 2698–2701. https://doi.org/10.1121/1.404385

[12] Robin Burchard, Pascal-André Brückner, Marius Bock, Juergen Gall, and Kristof Van Laerhoven. 2026. HARMES: A Multi-Modal Dataset for Wearable Human Activity Recognition with Motion, Environmental Sensing and Sound. *arXiv preprint arXiv:2605.02596* (2026).

[13] Shing Chan, Hang Yuan, Catherine Tong, Aidan Acquah, Abram Schonfeldt, Jonathan Gershuny, and Aiden Doherty. 2024. CAPTURE-24: A large dataset of wrist-worn activity tracker data collected in the wild for human activity recognition. *Scientific Data* 11, 1 (2024), 1135. https://doi.org/10.1038/s41597-024-03960-3

[14] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020. A Simple Framework for Contrastive Learning of Visual Representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*. PMLR, Virtual, 1597–1607.

[15] Ranak Roy Chowdhury, Ritvik Kapila, Ameya Panse, Xiyuan Zhang, Diyan Teng, Rashmi Kulkarni, Dezhi Hong, Rajesh K. Gupta, and Jingbo Shang. 2025. ZeroHAR: Sensor Context Augments Zero-Shot Wearable Action Recognition. In *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 39. AAAI Press, Philadelphia, PA, 16046–16054. https://doi.org/10.1609/aaai.v39i15.33762

[16] Utsav Dutta, Sina Khoshfetrat Pakazad, and Henrik Ohlsson. 2025. Time to Embed: Unlocking Foundation Models for Time Series with Channel Descriptions. *arXiv preprint arXiv:2505.14543* (2025). https://doi.org/10.48550/arXiv.2505.14543 System name: CHARM.

[17] Utsav Dutta, Gerardo Pastrana, Sina Khoshfetrat Pakazad, and Henrik Ohlsson. 2026. Giving Sensors a Voice: Multimodal JEPA for Semantic Time-Series Embeddings. In *Proceedings of the 43rd International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research)*, Vol. 306. PMLR, Seoul, South Korea. arXiv:2605.31580.

[18] Juntao Fang, Shifeng Xie, Shengbin Nie, Yuhui Ling, Yuming Liu, Zijian Li, Keli Zhang, Lujia Pan, Themis Palpanas, and Ruichu Cai. 2026. Rethinking Zero-Shot Time Series Classification: From Task-specific Classifiers to In-Context Inference. *arXiv preprint arXiv:2602.00620* (2026). System name: TIC-FM.

[19] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems 30 (NIPS)*. Curran Associates, Long Beach, CA, 4878–4887.

[20] Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. 2024. MOMENT: A Family of Open Time-Series Foundation Models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. PMLR, Vienna, Austria, 16115–16152.

[21] Lars Graf, Thomas Ortner, Stanisław Woźniak, and Angeliki Pantazi. 2026. FlowState: Sampling-Rate-Equivariant Time-Series Forecasting. In *Proceedings of the 43rd International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research)*, Vol. 306. PMLR, Seoul, South Korea. arXiv:2508.05287.

[22] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Ávila Pires, Zhaohan Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. 2020. Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*. Curran Associates, Virtual, 21271–21284.

[23] Zhiqing Hong, Zelong Li, Shuxin Zhong, Wenjun Lyu, Haotian Wang, Yi Ding, Tian He, and Desheng Zhang. 2024. CrossHAR: Generalizing Cross-dataset Human Activity Recognition via Hierarchical Self-Supervised Pretraining. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* 8, 2, Article 64 (2024), 26 pages. https://doi.org/10.1145/3659597

[24] Ahmet Iscen, Mathilde Caron, Alireza Fathi, and Cordelia Schmid. 2024. Retrieval-Enhanced Contrastive Vision-Text Models. In *The Twelfth International Conference on Learning Representations (ICLR)*. OpenReview.net, Vienna, Austria.

[25] Holger Junker, Oliver Amft, Paul Lukowicz, and Gerhard Tröster. 2008. Gesture spotting with body-worn inertial sensors to detect user activities. *Pattern Recognition* 41, 6 (2008), 2010–2024. https://doi.org/10.1016/j.patcog.2007.11.016

[26] Seyed Reza Kamel Tabbakh Farizani, Iman Naeimi, Kosar Naghavi, Fatemeh Majidi Nasab, Mehran Ghaffarian, and Mahdieh Sadat Sharifi Moghaddam Kakhki. 2026. InclusiveHAR: A Smartphone-Based Dataset for Human Activity Recognition Across Diverse Physical Abilities. Mendeley Data, V4. (2026). https://doi.org/10.17632/r78dn3f6nc.4

[27] Dahyun Kang, Ahmet Iscen, Eunchan Jo, Sua Choi, Minsu Cho, and Cordelia Schmid. 2025. Memory-Modular Classification: Learning to Generalize with Memory Replacement. *Transactions on Machine Learning Research (TMLR)* (2025). arXiv:2504.06021.

[28] Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. 2020. Generalization through Memorization: Nearest Neighbor Language Models. In *The Eighth International Conference on Learning Representations (ICLR)*. OpenReview.net, Addis Ababa, Ethiopia.

[29] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. 2020. Supervised Contrastive Learning. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*. Curran Associates, Virtual, 18661–18673.

[30] Kaushik Koneripalli, Suhas Lohit, Rushil Anirudh, and Pavan Turaga. 2020. Rate-Invariant Autoencoding of Time-Series. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Barcelona, Spain, 3732–3736. https://doi.org/10.1109/ICASSP40776.2020.9053983

[31] Jaris Küken, Shi Bin Hoo, Martin Mráz, Frank Hutter, and Lennart Purucker. 2026. TimEE: End-to-end Time Series Classification via In-Context Learning. *arXiv preprint arXiv:2607.07500* (2026).

[32] Bo Lan, Pei Li, Jiaxi Yin, Yunpeng Song, Ge Wang, Han Ding, Jinsong Han, and Fei Wang. 2025. XRF V2: A Dataset for Action Summarization with Wi-Fi Signals, and IMUs in Phones, Watches, Earbuds, and Glasses. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* 9, 3 (2025), 1–41. https://doi.org/10.1145/3749521

[33] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*. Curran Associates, Virtual, 9459–9474.

[34] Zechen Li, Baiyu Chen, Hao Xue, and Flora D. Salim. 2026. ZARA: Training-Free Motion Time-Series Reasoning via Evidence-Grounded LLM Agents. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics. arXiv:2508.04038.

[35] Zechen Li, Shohreh Deldari, Linyao Chen, Hao Xue, and Flora D. Salim. 2025. SensorLLM: Aligning Large Language Models with Motion Sensors for Human Activity Recognition. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Suzhou, China, 354–379. https://doi.org/10.18653/v1/2025.emnlp-main.19

[36] Jiayang Liu, Lin Zhong, Jehan Wickramasuriya, and Venu Vasudevan. 2009. uWave: Accelerometer-based personalized gesture recognition and its applications. *Pervasive and Mobile Computing* 5, 6 (2009), 657–675. https://doi.org/10.1016/j.pmcj.2009.07.007

[37] Aleksej Logacjov, Kerstin Bach, Atle Kongsvold, Hilde Bremseth Bårdstu, and Paul Jarle Mork. 2021. HARTH: A Human Activity Recognition Dataset for Machine Learning. *Sensors* 21, 23 (2021), 7853. https://doi.org/10.3390/s21237853

[38] Yunfei Luo, Yuliang Chen, Asif Salekin, and Tauhidur Rahman. 2026. Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals. *ACM Transactions on Computing for Healthcare* 7, 3 (2026), 1–43. https://doi.org/10.1145/3803808 Preprint: arXiv:2412.09758.

[39] Mohammad Malekzadeh, Richard G. Clegg, Andrea Cavallaro, and Hamed Haddadi. 2019. Mobile Sensor Data Anonymization. In *Proceedings of the International Conference on Internet of Things Design and Implementation (IoTDI)*. ACM, Montreal, Canada, 49–58.

[40] Miguel Matey-Sanz, Sven Casteleyn, and Carlos Granell. 2023. Dataset of Inertial Measurements of Smartphones and Smartwatches for Human Activity Recognition. *Data in Brief* 51 (2023), 109809. https://doi.org/10.1016/j.dib.2023.109809

[41] Conor McCarthy, Loes Quirijnen, Jan Peter van Zandwijk, Zeno Geradts, and Marcel Worring. 2025. Hi-OSCAR: Hierarchical Open-set Classifier for Human Activity Recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* 9, 4, Article 199 (2025), 27 pages. https://doi.org/10.1145/3770681

[42] Shenghuan Miao and Ling Chen. 2024. GOAT: A Generalized Cross-Dataset Activity Recognition Framework with Natural Language Supervision. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* 8, 4, Article 184 (2024), 28 pages. https://doi.org/10.1145/3699736

[43] Daniela Micucci, Marco Mobilio, and Paolo Napoletano. 2017. UniMiB SHAR: A Dataset for Human Activity Recognition Using Acceleration Data from Smartphones. *Applied Sciences* 7, 10 (2017), 1101.

[44] Kengo Nakata, Youyang Ng, Daisuke Miyashita, Asuka Maki, Yu-Chieh Lin, and Jun Deguchi. 2022. Revisiting a kNN-Based Image Classification System with High-Capacity Storage. In *Computer Vision – ECCV 2022, 17th European Conference, Proceedings, Part XXXVII (Lecture Notes in Computer Science)*, Vol. 13697. Springer, Tel Aviv, Israel, 457–474. https://doi.org/10.1007/978-3-031-19836-6_26

[45] Girish Narayanswamy, Xin Liu, Kumar Ayush, Yuzhe Yang, Xuhai Xu, Shun Liao, Jake Garrison, Shyam A. Tailor, Jacob E. Sunshine, Yun Liu, Tim Althoff, Shrikanth Narayanan, Pushmeet Kohli, Jiening Zhan, Mark Malhotra, Shwetak N. Patel, Samy Abdel-Ghaffar, and Daniel McDuff. 2025. Scaling Wearable Foundation Models. In *The Thirteenth International Conference on Learning Representations (ICLR)*. OpenReview.net, Singapore. arXiv:2410.13638.

[46] Long-Van Nguyen-Dinh, Alberto Calatroni, and Gerhard Tröster. 2014. Robust Online Gesture Recognition with Crowdsourced Annotations. *Journal of Machine Learning Research (JMLR)* 15, 1 (2014), 3187–3220.

[47] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2023. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *The Eleventh International Conference on Learning Representations (ICLR)*. OpenReview.net, Kigali, Rwanda.

[48] Mohammad Norouzi, Tomas Mikolov, Samy Bengio, Yoram Singer, Jonathon Shlens, Andrea Frome, Greg S. Corrado, and Jeffrey Dean. 2014. Zero-Shot Learning by Convex Combination of Semantic Embeddings. In *International Conference on Learning Representations (ICLR)*. arXiv:1312.5650.

[49] Francisco Javier Ordóñez and Daniel Roggen. 2016. Deep Convolutional and LSTM Recurrent Neural Networks for Multimodal Wearable Activity Recognition. *Sensors* 16, 1 (2016), 115.

[50] Xiaomin Ouyang, Xian Shuai, Jiayu Zhou, Ivy Wang Shi, Zhiyuan Xie, Guoliang Xing, and Jianwei Huang. 2022. Cosmo: Contrastive Fusion Learning with Small Data for Multimodal Human Activity Recognition. In *Proceedings of the 28th Annual International Conference on Mobile Computing and Networking (MobiCom)*. ACM, 324–337.

[51] Xiaomin Ouyang, Zhiyuan Xie, Jiayu Zhou, Jianwei Huang, and Guoliang Xing. 2021. ClusterFL: A Similarity-Aware Federated Learning System for Human Activity Recognition. In *Proceedings of the 19th Annual International Conference on Mobile Systems, Applications, and Services (ACM MobiSys)*. ACM, 54–66. https://doi.org/10.1145/3458864.3467681

[52] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*. PMLR, Virtual, 8748–8763.

[53] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Hong Kong, China, 3982–3992.

[54] Attila Reiss and Didier Stricker. 2012. Introducing a New Benchmarked Dataset for Activity Monitoring. In *Proceedings of the 16th International Symposium on Wearable Computers (ISWC)*. IEEE, Newcastle, UK, 108–109.

[55] Jorge-Luis Reyes-Ortiz, Luca Oneto, Albert Samà, Xavier Parra, and Davide Anguita. 2016. Transition-Aware Human Activity Recognition Using Smartphones. *Neurocomputing* 171 (2016), 754–767.

[56] Daniel Roggen, Alberto Calatroni, Mirco Rossi, Thomas Holleczek, Kilian Förster, Gerhard Tröster, Paul Lukowicz, David Bannach, Gerald Pirkl, Alois Ferscha, et al. 2010. Collecting Complex Activity Datasets in Highly Rich Networked Sensor Environments. In *Proceedings of the 7th International Conference on Networked Sensing Systems (INSS)*. IEEE, 233–240.

[57] Murat Sensoy, Lance Kaplan, and Melih Kandemir. 2018. Evidential Deep Learning to Quantify Classification Uncertainty. In *Advances in Neural Information Processing Systems 31 (NeurIPS)*. Curran Associates, Montreal, Canada, 3183–3193.

[58] Jeff Shen, Francois Lanusse, Liam Holden Parker, Ollie Liu, Tom Hehir, Leopoldo Sarra, Lucas Meyer, Micah Bowles, Sebastian Wagner-Carena, Helen Qu, Siavash Golkar, Alberto Bietti, Kyunghyun Cho, Miles Cranmer, and Shirley Ho. 2025. Universal Spectral Tokenization via Self-Supervised Panchromatic Representation Learning. In *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences (ML4PS)*. San Diego, CA. arXiv:2510.17959.

[59] Muhammad Shoaib, Stephan Bosch, Ozlem Durmaz Incel, Hans Scholten, and Paul J. M. Havinga. 2014. Fusion of Smartphone Motion Sensors for Physical Activity Recognition. *Sensors* 14, 6 (2014), 10146–10176.

[60] Muhammad Shoaib, Stephan Bosch, Ozlem Durmaz Incel, Hans Scholten, and Paul J. M. Havinga. 2016. Complex Human Activity Recognition Using Smartphone and Wrist-Worn Motion Sensors. *Sensors* 16, 4 (2016), 426. https://doi.org/10.3390/s16040426

[61] Niloy Sikder and Abdullah-Al Nahid. 2021. KU-HAR: An Open Dataset for Heterogeneous Human Activity Recognition. *Pattern Recognition Letters* 146 (2021), 46–54. https://doi.org/10.1016/j.patrec.2021.02.024

[62] Nirhoshan Sivaroopan, Hansi Karunarathna, Chamara Madarasingha, Anura Jayasumana, and Kanchana Thilakarathna. 2026. RAG-HAR: Retrieval Augmented Generation-based Human Activity Recognition. In *Proceedings of the IEEE International Conference on Pervasive Computing and Communications (PerCom)*. IEEE. arXiv:2512.08984.

[63] Jake Snell, Kevin Swersky, and Richard Zemel. 2017. Prototypical Networks for Few-shot Learning. In *Advances in Neural Information Processing Systems 30 (NIPS)*. Curran Associates, Long Beach, CA, 4077–4087.

[64] Allan Stisen, Henrik Blunck, Sourav Bhattacharya, Thor Siiger Prentow, Mikkel Bent Kjærgaard, Anind Dey, Tobias Sonne, and Mads Møller Jensen. 2015. Smart Devices Are Different: Assessing and

Mitigating Mobile Sensing Heterogeneities for Activity Recognition. In *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems (SenSys)*. ACM, Seoul, South Korea, 127–140.

[65] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. 2024. RoFormer: Enhanced Transformer with Rotary Position Embedding. *Neurocomputing* 568 (2024), 127063. https://doi.org/10.1016/j.neucom.2023.127063

[66] Timo Sztyler and Heiner Stuckenschmidt. 2016. On-Body Localization of Wearable Devices: An Investigation of Position-Aware Activity Recognition. In *Proceedings of the IEEE International Conference on Pervasive Computing and Communications (PerCom)*. IEEE, Sydney, Australia, 1–9.

[67] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*. Curran Associates, Long Beach, CA, 5998–6008.

[68] George Vavoulas, Charikleia Chatzaki, Thodoris Malliotakis, Matthew Pediaditis, and Manolis Tsiknakis. 2016. The MobiAct Dataset: Recognition of Activities of Daily Living Using Smartphones. In *Proceedings of the International Conference on Information and Communication Technologies for Ageing Well and e-Health (ICT4AWE)*. SciTePress, Rome, Italy, 143–151.

[69] Oriol Vinyals, Charles Blundell, Timothy Lillicrap, Koray Kavukcuoglu, and Daan Wierstra. 2016. Matching Networks for One Shot Learning. In *Advances in Neural Information Processing Systems 29 (NIPS)*. Curran Associates, Barcelona, Spain, 3630–3638.

[70] Qingxin Wei, Jiaming Huang, Yi Gao, and Wei Dong. 2025. One Model to Fit Them All: Universal IMU-based Human Activity Recognition with LLM-assisted Cross-dataset Representation. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* 9, 3, Article 139 (2025), 22 pages. https://doi.org/10.1145/3749509

[71] Gary M. Weiss, Kenichi Yoneda, and Thaier Hayajneh. 2019. Smartphone and Smartwatch-Based Biometrics Using Activities of Daily Living. *IEEE Access* 7 (2019), 133190–133202. https://doi.org/10.1109/ACCESS.2019.2940729

[72] Yongqin Xian, Christoph H. Lampert, Bernt Schiele, and Zeynep Akata. 2019. Zero-Shot Learning—A Comprehensive Evaluation of the Good, the Bad and the Ugly. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41, 9 (2019), 2251–2265. https://doi.org/10.1109/TPAMI.2018.2857768

[73] Huatao Xu, Pengfei Zhou, Rui Tan, and Mo Li. 2023. Practically Adopting Human Activity Recognition. In *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking (ACM MobiCom)*. ACM, Article 85, 15 pages. https://doi.org/10.1145/3570361.3613299

[74] Huatao Xu, Pengfei Zhou, Rui Tan, Mo Li, and Guobin Shen. 2021. LIMU-BERT: Unleashing the Potential of Unlabeled Data for IMU Sensing Applications. In *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems (SenSys)*. ACM, Coimbra, Portugal, 220–233.

[75] Maxwell A. Xu, Jaya Narain, Gregory Darnell, Haraldur Tómas Hallgrímsson, Hyewon Jeong, Darren Forde, Richard Fineman, Karthik Jayaraman Raghuram, James M. Rehg, and Shirley You Ren. 2025. RelCon: Relative Contrastive Learning for a Motion Foundation Model for Wearable Data. In *The Thirteenth International Conference on Learning Representations (ICLR)*. OpenReview.net, Singapore. arXiv:2411.18822.

[76] Maxwell A. Xu, Girish Narayanswamy, Kumar Ayush, Dimitris Spathis, Shun Liao, Shyam A. Tailor, Ahmed Metwally, A. Ali Heydari, Yuwei Zhang, Jake Garrison, Samy Abdel-Ghaffar, Xuhai Xu, Ken Gu, Jacob E. Sunshine, Ming-Zher Poh, Yun Liu, Tim Althoff, Shrikanth Narayanan, Pushmeet Kohli, Mark Malhotra, Shwetak N. Patel, Yuzhe Yang, James M. Rehg, Xin Liu, and Daniel McDuff. 2025. LSM-2: Learning from Incomplete Wearable Sensor Data. *arXiv preprint arXiv:2506.05321* (2025). https://doi.org/10.48550/arXiv.2506.05321

[77] Zongzhe Xu, Aakarsh Anand, Sarah Jiang, Chuntung Zhuang, Zitao Shuai, Sriram Sankararaman, and Yuzhe Yang. 2026. Inertia-1: An Open Exploration of Wearable Motion Foundation Models. *arXiv preprint arXiv:2607.06617* (2026).

[78] Hua Yan, Heng Tan, Yi Ding, Pengfei Zhou, Vinod Namboodiri, and Yu Yang. 2025. Large Language Model-guided Semantic Alignment for Human Activity Recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* 9, 4 (2025), 1–25. https://doi.org/10.1145/3770652 System name: LanHAR.

[79] Yan Yan, Dali Chen, Yushi Liu, Jinjin Zhao, Bo Wang, Xuankun Wu, Xiaohao Jiao, Yuqian Chen, Huihui Li, and Xuchao Ren. 2021. TNDA-HAR. IEEE DataPort. (2021). https://doi.org/10.21227/4epb-pg26

[80] Hang Yuan, Shing Chan, Andrew P. Creagh, Catherine Tong, Aidan Acquah, David A. Clifton, and Aiden Doherty. 2024. Self-supervised learning for human activity recognition using 700,000 person-days of wearable data. *npj Digital Medicine* 7, 1, Article 91 (2024), 10 pages. https://doi.org/10.1038/s41746-024-01062-3 arXiv:2206.02909.

[81] Hao Zhang, Zhan Zhuang, Xuehao Wang, Xiaodong Yang, and Yu Zhang. 2025. MoPFormer: Motion-Primitive Transformer for Wearable-Sensor Activity Recognition. In *Advances in Neural Information Processing Systems 38 (NeurIPS)*. Curran Associates, San Diego, CA. arXiv:2505.20744.

[82] Mi Zhang and Alexander A. Sawchuk. 2012. USC-HAD: A Daily Activity Dataset for Ubiquitous Activity Recognition Using Wearable Sensors. In *Proceedings of the 2012 ACM Conference on Ubiquitous Computing (UbiComp)*. ACM, Pittsburgh, PA, 1036–1043. https://doi.org/10.1145/2370216.2370438

[83] Xiyuan Zhang, Danielle C. Maddix, Junming Yin, Nick Erickson, Abdul Fatir Ansari, Boran Han, Shuai Zhang, Leman Akoglu, Christos Faloutsos, Michael W. Mahoney, Cuixiong Hu, Huzefa Rangwala, George Karypis, and Bernie Wang. 2025. Mitra: Mixed Synthetic Priors for Enhancing Tabular Foundation Models. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2510.21204.

[84] Xiyuan Zhang, Diyan Teng, Ranak Roy Chowdhury, Shuheng Li, Dezhi Hong, Rajesh K. Gupta, and Jingbo Shang. 2024. UniMTS: Unified Pre-training for Motion Time Series. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.

[85] Xiang Zhang, Ziyuan Zhao, Theodoros Tsiligkaridis, and Marinka Zitnik. 2022. Self-Supervised Contrastive Pre-Training for Time Series via Time-Frequency Consistency. In *Advances in Neural Information Processing Systems 35 (NeurIPS)*. Curran Associates, New Orleans, LA, 3988–4003.

[86] Yuwei Zhang, Kumar Ayush, Siyuan Qiao, A. Ali Heydari, Girish Narayanswamy, Maxwell A. Xu, Ahmed A. Metwally, Shawn Xu, Jake Garrison, Xuhai Xu, Tim Althoff, Yun Liu, Pushmeet Kohli, Jiening Zhan, Mark Malhotra, Shwetak Patel, Cecilia Mascolo, Xin Liu, Daniel McDuff, and Yuzhe Yang. 2025. SensorLM: Learning the Language of Wearable Sensors. In *Advances in Neural Information Processing Systems 38 (NeurIPS)*. Curran Associates, San Diego, CA. arXiv:2506.09108.

[87] Shibo Zhao, Sifan Zhou, Raphael Blanchard, Yuheng Qiu, Wenshan Wang, and Sebastian Scherer. 2025. Tartan IMU: A Light Foundation Model for Inertial Positioning in Robotics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

A THE DEPENDENCY STRUCTURE BEHIND THE COMBINATION

Section 3.3 claims HALO's four choices are *mutually entailed* rather than independently attractive. Table 7 records which axes the closest systems claim to address; the four dependencies follow in full.

Table 7: Which axes each system *explicitly claims* to address. ✓ = in the form we argue is needed; ○ = a weaker form (rate *equivariance* rather than physical-Hz anchoring; *categorical* rather than free-text placement; rotation *invariance*, which discards gravity). The last column marks systems reporting a range of label budgets with no gradient update. Blank is not a criticism—these are different papers solving different problems—and the table reports stated scope, not measured performance. The point is not the bottom row's length but the dependency structure below.

	acquisition		orient./	label	k-curve
System	rate	place	grav.	interf.	
LiMU-BERT [74], CrossHAR [23]					
harnet [80], LSM [45, 76]					✓
NormWear [38]		○		✓	
UniMTS [84]		✓	○	✓	
ZeroHAR [15]		○		✓	
GOAT [42], oneHAR [70]		✓		✓	
CHARM [16, 17]		✓			
MoPFormer [81]		○			
FlowState [21]	○				
RAG-HAR [62], ZARA [34]				✓	✓
SensorLM [86]				✓	✓
HALO	✓	✓	✓	✓	✓

Rate anchoring is a precondition for the memory, not a parallel feature. A non-parametric memory only works if a query and a stored exemplar recorded on different devices are comparable *as vectors*. If the encoder is rate-*equivariant*—told the rate and adapting—then two windows at different rates are related by a learned transformation, and the memory must either be indexed per configuration or trust that transformation to be exact. Because HALO's band k denotes the same physical frequency for every device, one flat memory is well-posed. L1 exists in the form it does because of Phase B.

The granularity of the language conditioning is forced by the retrieval. Per-channel free text [16, 17, 35] is the natural choice if the conditioning feeds a classifier: it is maximally expressive and the head can ignore what it does not need. It is the wrong choice if the conditioning feeds a memory, because a description nearly unique to a corpus makes *same-corpus* neighbours the nearest ones—precisely the shortcut cross-dataset evaluation exists to catch. Attaching identity to the *sensor* keeps the description expressive enough to name an unseen device and too coarse to name a dataset. That is why the dataset-identity probe is in the paper.

Preserving gravity is forced by the geometry finding. Rotation invariance [84] is the standard answer to unknown mounting, and it is lossy in one specific way: it discards the gravity direction, and with it posture. Section 3.3 shows that the label pairs a text projection cannot separate are exactly the posture-versus- transition pairs (*sitting*/*sitting down*, *lying*/*standing*). A system that gives up on the text geometry *must* therefore keep posture in the signal, so conditioning rather than invariance is not a preference here but a requirement.

Label-free pretraining is forced by the same shortcut argument. Any closed-vocabulary objective over a multi-corpus training set rewards inferring the corpus and predicting its marginal. That is survivable for a parametric head, which can be refitted per deployment; it is not survivable for a frozen encoder feeding a memory, because the shortcut is baked into the stored vectors.

So the four choices form a chain: removing any one invalidates another. That is a different and more falsifiable claim than "we do all of these," and Section 5.4 is built to test it—each ablation in Table 6 removes one link and asks whether the rest still stand.

B CORPUS

Table 2 in the body lists the training corpus and the held-out evaluation sets; Figure 5 plots them by placement and native rate. Window counts there are windows on the native-rate grid after physical-plausibility and duplicate filtering. Most windows are six seconds, but UCI-HAR ships 2.56 s records and SP-SW-HAR 1 s, so hours are computed from each grid's own samples and rate rather than by multiplying by six.

Deliberate exclusions. Six datasets were excluded from both splits. DSADS [6], HARTH [37], and Opportunity [56] use torso and limb IMU harnesses rather than phone or watch placements. RecGym [8] is distributed per-axis min–max normalised, which destroys the physical scale and gravity component the tokenizer depends on. MobiAct [68] was never materialised into grids. And HAPT [55] is excluded because it re-records the same 30 subjects as UCI-HAR [1], so including both would place the same people on both sides of a supposedly subject-disjoint boundary.

C BASELINES

Table 8 lists the five baselines, what each was pretrained on, and how much acquisition context each can receive at inference—the axis on which the comparison is not symmetric.

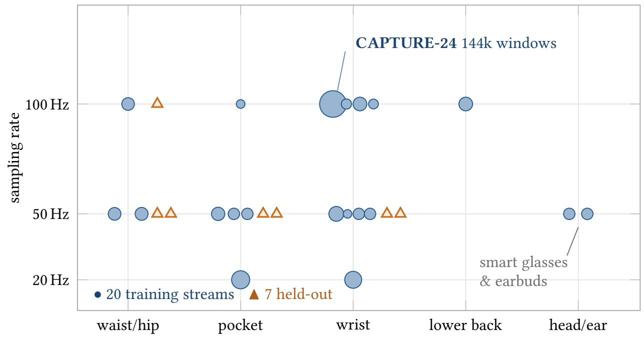

Figure 5: The corpus does not cover a grid; it covers a scatter. Each blue disc is one of the 20 sensor streams HALO trains on, positioned by body placement and native sampling rate, with area proportional to the number of 6 s windows it contributes; orange triangles are the 7 held-out evaluation datasets. Rates span a 5× range and placements include two—smart glasses and wireless earbuds—that no established HAR benchmark contains. Held-out cells are not simply new subjects: they combine placements and rates in proportions the training corpus never sees. Absorbing this by resampling to a common rate would discard the content above 10 Hz that only the 50/100 Hz streams can resolve.

Table 8: Baselines and where each sits on the heterogeneity stack. "Blind" means the model receives no information about where the sensor was worn. Two baselines are pretrained by us on our corpus with their published objectives (data-matched); three are used as released (faithful, but pretrained on their own, larger corpora). We report the two groups separately rather than averaging over them.

Model	Weights	Rate	Channels / placement	Labels
LiMU-BERT [74]	we pretrain	20 Hz	6-ch pad+mask; blind	ConSE
CrossHAR [23]	we pretrain	20 Hz	6-ch pad+mask; blind	ConSE
harnet [80]	released	30 Hz	3-ch acc; wrist	ConSE
UniMTS [84]	released	resamp.	joint-graph placement	native
NormWear [38]	released	65 Hz	variable ch; blind	native
HALO (ours)	we train	**native**	**per-sensor language**	**native**

D PARITY CONTROL

Because several baselines cannot accept HALO's native inputs, Table 9 reports HALO restricted to each baseline's own input contract, so any remaining gap is attributable to the model rather than to what it was allowed to read.

E GRAVITY AS A SIGNED DC FEATURE

Section 4.2.1 keeps the quasi-static component as a signed per-axis feature rather than normalising it away. Figure 7 is the measurement behind that choice.

Table 9: Parity control: HALO under its native inputs versus baseline-equivalent inputs. Baselines are split by whether their weights were trained on our corpus or released by their authors.

Configuration	k=0	k=10
HALO (native Hz, per-sensor text)		
HALO *parity* (20 Hz, neutral text)		
Best baseline (data-matched group)		
Best baseline (released-checkpoint group)		

F TOKENIZER GUARDRAILS

Zero-padding beyond the true patch length is frequency-domain interpolation only and does not alter band energies; we verified that S=256 and S=512 give band-energy cosine 1.000000 and identical masks at 20, 25, 50, and 100 Hz. Patch lengths exceeding S raise an error rather than silently truncating.

G PHASE-A SPECIFICATION

The EMA teacher is updated only after an optimizer step (decay 0.996) and is checkpointed with the model. Cross-placement positives are drawn under a quota so that the six-placement recordings cannot dominate; positive identity comes from explicit physical-event identifiers written during conversion and carried into the pretraining index, never inferred from equal array lengths or row alignment, and never from an activity label. Per-objective gradient norms, VICReg component values, minimum feature standard deviation, effective rank, and positive similarity by pair type are logged every step; non-finite values or representation collapse fail the run rather than being smoothed over.

H PHASE-A OBJECTIVE ABLATION

Table 10: Phase-A objective ablation, macro-F1 over the held-out datasets. The last two rows are controls, not variants: they test whether the non-contrastive and label-free choices were worth making.

Phase-A configuration	k=0	k=10
Full (A1 fixed+EMA, VICReg, TF, placement, rail)		
− EMA-latent target (fixed A1 only)		
− augmentation VICReg		
− time–frequency consistency		
− cross-placement positives		
− physical grounding rail		
VICReg ⟶ NT-Xent (contrastive control)		
Label-supervised recipe (SupCon; *uses labels*)		

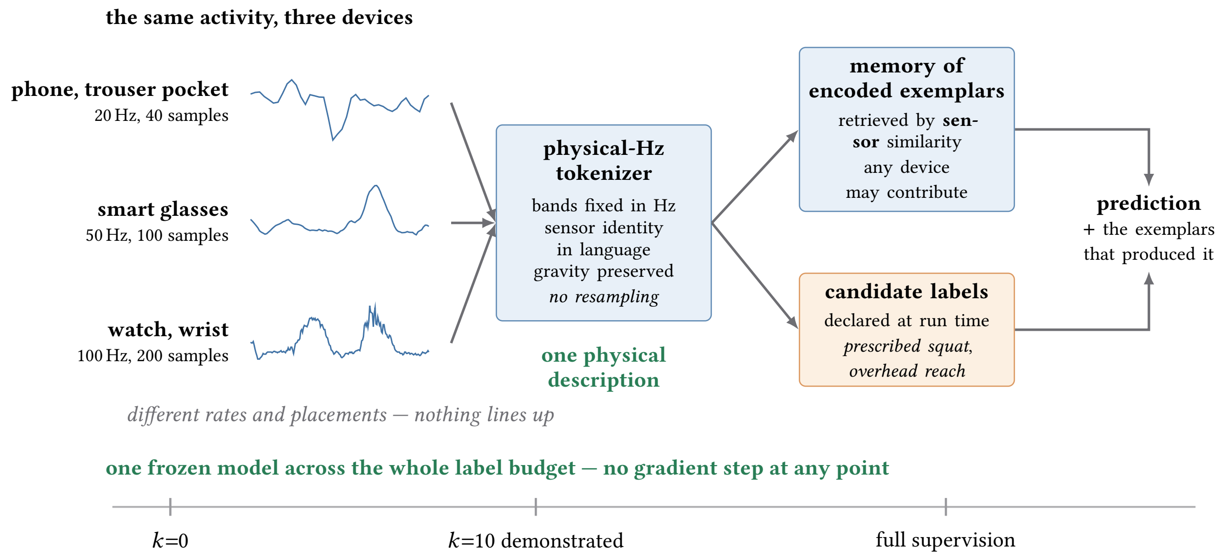

Figure 6: What "no closed grammar" looks like in the corpus. The three traces on the left are real accelerometer snippets of *walking*, two seconds each, from three streams of the materialised corpus; they are amplitude-normalised for display, but the sample counts are the true ones. As arrays they have nothing in common. HALO absorbs that heterogeneity *in the tokenizer*—a frequency axis fixed in Hz, sensor identity supplied in language, gravity preserved—so windows recorded on different devices become directly comparable as vectors, which is the precondition for one flat memory. Recognition is then a retrieval in *sensor* space, with the label strings a deployment declares at run time used only to name and to enumerate; because the memory is append-only, one frozen model spans the whole label budget with no gradient update at any point.

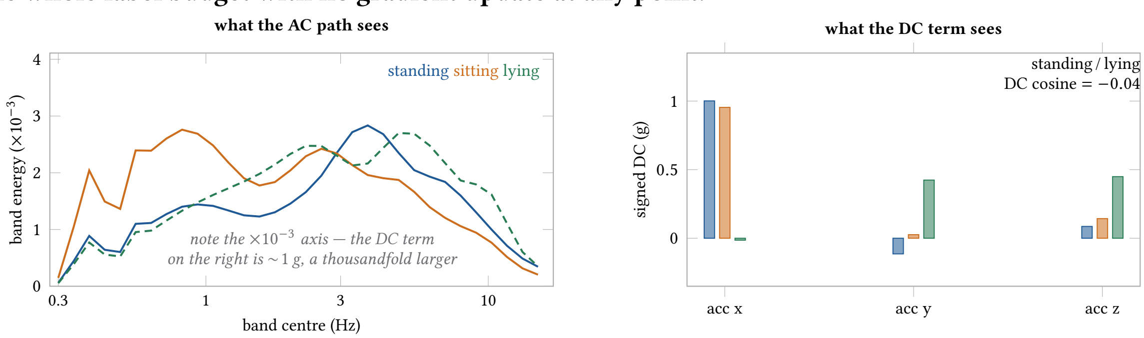

Figure 7: Why gravity is conditioned on rather than normalised away. Three static postures, UCI-HAR waist phone, 200 windows each through the real tokenizer path. *Left:* the frequency path receives almost nothing (note the $\times 10^{-3}$ axis) and separates no pair distinctly—all pairwise cosines fall within 0.84–0.88. *Right:* signed DC places lying nearly orthogonal to standing (−0.04), because gravity has redistributed across axes. The limit is shown too: standing and sitting stay at 0.99 in DC, the same pair Section 3.3 finds inseparable in text.

I PHASE-B EPISODIC TRAINING

The decoder is trained on episodes that vary, independently: the candidate budget (10–100% of the vocabulary); the number of true-label memory exemplars (0, 1, 2, 4, 8, or all); the support available for competing labels; whether memory is same-configuration, cross-configuration only, or missing the query's configuration entirely; the kind of distractors present (random, language-near, motion-family-near, physically confusable); and whether the truth is present at all. Complete label *families* are held out of decoder training and reserved for model selection, so candidate-text reasoning is always tested on concepts the decoder never fit. Head collapse in the retrieval projections is resisted by output decorrelation, projection diversity, and contribution-usage regularisation.

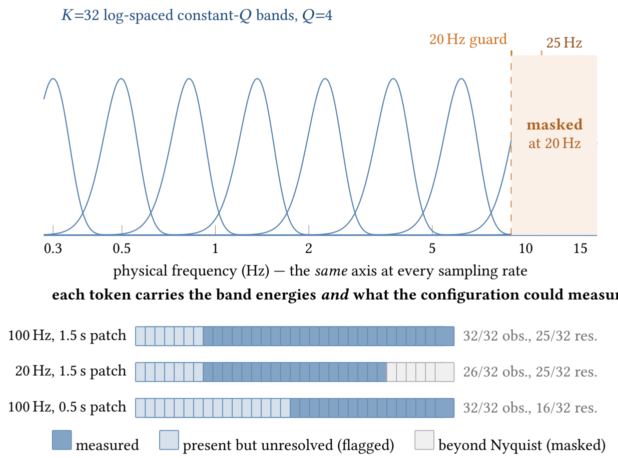

Figure 8: The filterbank, and what a token carries. ***Top:*** **K=32 Gaussian bands log-spaced over 0.3–15 Hz; because a native-rate zero-padded rDFT maps bin m to mr/S, one bank reads every rate without resampling. Dashed lines mark the Nyquist guard for the two lowest corpus rates.** ***Bottom:*** **the real observability and resolution vectors emitted alongside the energies—part of the token, not diagnostics.**